

Emotion agent: Unsupervised deep reinforcement learning with distribution-prototype reward for continuous emotional EEG analysis

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ABSTRACT

Continuous Electroencephalography (EEG) signals are widely employed in affective brain–computer interface (aBCI) applications. However, only a subset of the continuously acquired EEG data is truly relevant to emotional processing, while the remainder is often noisy or unrelated. Manual annotation of these key emotional segments is impractical due to their dynamic and individualized nature. To address this challenge, we propose a novel unsupervised deep reinforcement learning framework, termed *Emotion Agent*, which automatically identifies and extracts the most informative emotional segments from continuous EEG signals. Emotion Agent initially utilizes a heuristic algorithm to perform a global search and generate prototype representations of the EEG signals. These prototypes guide the exploration of the signal space and highlight regions of interest. Furthermore, we design a distribution-prototype-based reward function that evaluates the interaction between samples and prototypes to ensure that the selected segments are both representative and relevant to the underlying emotional states. Finally, the framework is trained using Proximal Policy Optimization (PPO) to achieve stable and efficient convergence. Experimental results on three widely used datasets (covering both discrete and dimensional emotion recognition) show an average improvement of 13.46 % when using the proposed Emotion Agent, demonstrating its significant enhancement of accuracy and robustness in downstream aBCI tasks.

1. Introduction

Human emotion is inherently dynamic, arising from complex interactions between internal cognitive processes and external stimuli [1,2]. Electroencephalography (EEG) provides an objective, direct method to assess these emotional states [3–6]. However, conventional approaches often treat all EEG samples equally, neglecting the fact that only specific segments truly capture the emotional dynamics [7,8]. This indiscriminate inclusion not only fails to reflect the evolving nature of human emotions but also introduces noise and confounding factors into the analysis [9–11]. Furthermore, real-time annotation of emotion-related segments is unfeasible since individuals cannot consistently distinguish emotional shifts. To overcome these issues, we reformulate the extraction of key emotional segments as a sequential decision-making problem and leverage deep reinforcement learning (DRL).

Deep reinforcement learning enables agents to autonomously explore and extract task-relevant features through reward-driven mechanisms, eliminating the need for extensive labeled data. For example, Zhou et al. [12] guided an agent to generate diverse and representative video summaries by introducing a diversity-representativeness reward, while Apostolidis et al. [13] employed an Actor-Critic framework to extract video key segments based on reconstruction error. In the EEG domain, Zhang et al. [11] demonstrated the potential of DRL with TAS-Net for extracting emotional segments. However, existing methods often overlook the global distribution of EEG features and struggle to capture long-term temporal dependencies. This limitation hinders a comprehensive understanding of the continuity and subtle transitions in emotional states. To address this, integrating reinforcement learning strategies that simultaneously account for short-term rewards and the continuity of

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long-term emotional dynamics may substantially improve the precision and effectiveness of emotion recognition models.

In this study, we reformulate the extraction of key EEG segments as a sequential decision-making process. We propose a novel framework, termed **Emotion Agent**, designed to automatically identify critical emotional moments from continuous EEG signals in an unsupervised manner, without the need for predefined labels. Emotion Agent synergistically combines the strengths of deep reinforcement learning with heuristic search algorithms. This hybrid approach enhances learning efficiency by guiding the agent toward high-reward trajectories while minimizing time spent on less informative exploration paths. To further improve the decision-making process, we introduce a distribution-prototype-based reward function that captures both local and global statistical characteristics of EEG features. This reward formulation ensures that the agent not only extracts the most emotionally significant segments but also preserves the intrinsic data distribution patterns. As a result, the model achieves faster and more stable convergence, even under the inherent exploration–exploitation trade-off. Our main contributions are summarized as follows:

- We propose Emotion Agent, an unsupervised framework that integrates heuristic search and deep reinforcement learning to automatically identify emotionally relevant EEG segments.
- We design a novel distribution-prototype reward function that leverages both local and global feature distribution characteristics, enabling more informed and robust agent behavior.
- Extensive experiments on the SEED, SEED-IV, and DEAP datasets (covering both discrete and dimensional emotion recognition tasks) demonstrate that Emotion Agent significantly enhances recognition accuracy and robustness across diverse settings.

2. Related work

2.1. Reinforcement learning

Reinforcement Learning (RL) is a powerful paradigm in which an agent learns to make optimal decisions by interacting with an environment and receiving rewards [14]. Unlike static methods, RL continuously adapts its actions to maximize cumulative rewards, making it well-suited for dynamic and complex environments [15,16].

2.2. Deep reinforcement learning in EEG

Deep Reinforcement Learning offers significant promise for automating and enhancing EEG-based BCI tasks like emotion recognition, sleep staging and action imagery.

To address the dynamic nature of EEG-based emotion recognition, Zhang et al. [11] proposed TAS-Net, an unsupervised time-aware sampling network via DRL. In contrast to traditional methods that rely on static, trial-level labels and ignore temporal dynamics, TAS-Net explicitly captured both short-term emotional continuity and long-range temporal dependencies. By identifying key emotional segments from continuous EEG signals, it enabled more informative feature extraction for downstream emotion recognition tasks. Experimental results showed that TAS-Net consistently outperforms existing unsupervised methods, offering a robust and effective solution for emotion-based BCI applications in dynamic and unlabeled environments.

To address the challenge of optimizing model structures, Yang et al. [17] introduced a hierarchical DQN-PSO framework. In this approach, a Deep Q-Network (DQN) selected the core CNN architecture, while Particle Swarm Optimization (PSO) fine-tuned convolutional kernel parameters. This hybrid strategy enabled automated generation of high-performing EEG classification models. To address the limitations of CNNs in capturing the non-Euclidean spatial structure of EEG electrodes, Aung et al. [18] proposed EEG RL-Net. This framework combined pretrained Graph Neural Networks (GNNs) for spatial feature extraction with a Dueling DQN module that dynamically skipped ambiguous temporal segments. The model improved motor imagery performance

and supported real-time applications. For applications involving limited clinical data, Liu et al. [19] developed AFM-DQN, a framework that incorporated Semi-Joint Mutual Information (Semi-JMI) for feature selection within a DQN-based pipeline. The model also integrated a pretrained module, SVM classifier, and reward control mechanism to facilitate effective localization of epileptogenic zones.

To overcome the constraints of explicit user feedback in reinforcement learning, EEG-derived Error-Related Potentials (ErrPs) are incorporated as implicit, low-latency reinforcement signals. For example, Xu et al. [20] demonstrated how ErrP-based feedback can reduce cognitive load and improve sample efficiency in real-time applications such as autonomous driving. Similarly, Luo et al. [21] embedded a CNN-decoded ErrP preference signal within a reinforcement teacher framework to enhance training in robotic systems.

Inspired by neuroscience, Li et al. [22] proposed FLD3QN, a biologically inspired DRL architecture that models emotional processing based on the Papez circuit and dopamine-driven reward mechanisms. This neurobiological grounding enabled significant improvements in emotion recognition accuracy, highlighting the potential of neuroscience-informed DRL designs in affective computing.

2.3. Reinforcement learning with heuristics

To accelerate traditional RL approaches, researchers have incorporated domain-specific heuristics to guide the exploration process, forming Heuristic-Guided Reinforcement Learning (HuRL) [23]. By incorporating heuristics derived from expert knowledge or offline data, HuRL guides agents toward more promising regions of the state space, thereby reducing the computational expense associated with unguided exploration. Recent advances in large-state space RL for hyper-heuristics have leveraged extensive feature sets and local search principles within the ϵ -greedy framework [24]. This method enhances the likelihood of sampling high-quality sequences of low-level heuristics, thereby improving the efficiency of RL in tackling complex optimization problems with vast state spaces.

Beyond these developments, further advancements underscore the versatility of reinforcement learning. For example, Jiang et al. introduced a two-stage Generative Adversarial Network (GAN) with A* heuristic search algorithm for continuous trajectory generation in urban road networks [25]. In this approach, discriminators evaluated both sequential and movement yaw rewards, guiding the agent to generate more accurate and efficient trajectories. Furthermore, recent frameworks in personalized reinforcement learning leveraged clustering techniques (such as K-means) and integrated a policy budget within robust Markov Decision Processes (r-MDPs) [26]. This enables RL agents to better adapt to individual user preferences through representative policies. Additionally, AgentX was developed to enhance the effectiveness of Intelligent Tutoring Systems (ITS) in the field of education, demonstrating RL's potential to tailor learning experiences by clustering personalized student data [27].

Advancements in reinforcement learning across various domains, such as accelerating traditional algorithms with heuristics, personalizing user interactions, and enhancing intelligent tutoring systems, demonstrate the superior versatility and potential of this paradigm. The continuous evolution of RL, exemplified by advancements in large-state optimization and urban trajectory generation, underscores its capacity to address increasingly complex challenges. As RL further integrates with complementary machine learning and deep learning techniques like GANs and heuristic search algorithms, it is well-positioned to drive significant improvements across diverse fields, ultimately shaping the future of intelligent systems and autonomous decision-making [28–30].

3. Methodology

3.1. Markov decision process formulation

To detect the key emotional segments from sequential EEG signals, we formulate the task as a sequential decision-making problem modeled

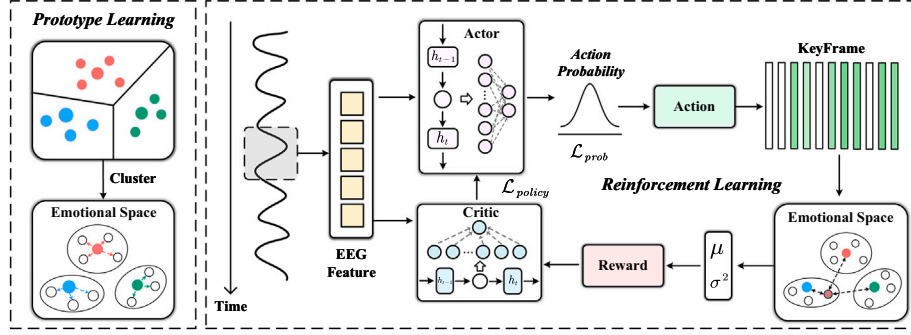


Fig. 1. The proposed emotion agent framework: heuristic prototype initialization and prototype-guided reinforcement learning.

using a Markov Decision Process (MDP). An MDP is defined as a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. Here, the state space $\mathcal{S}$ represents various segments of the EEG signals, and the action space $\mathcal{A} = \{0, 1\}$ consists of two possible actions (action 1 indicates the selection of a segment as emotionally relevant, and action 0 denotes its rejection). The transition probability function $\mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ describes the likelihood of transitioning from one state to another given a specific action. $\gamma \in [0, 1]$ is the discount factor, reflecting the agent's trade-off between short-term and long-term interests. The reward function $\mathcal{R} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathcal{R}$ assigns a numerical reward based on the state-action-state transition, providing feedback on the agent's decisions.

Within an MDP, at each timestep t , the reinforcement learning agent begins by observing the current state s_t , then selects an action a_t , and finally transitions to a new state s_{t+1} while receiving an immediate reward r_t . The resulting sequence of state-action transitions forms a trajectory, and as the agent repeats this process over successive iterations, it collects multiple trajectories of experience. Over time, this continuous interaction enables the agent to learn an optimal policy, denoted by π^* , that maximizes cumulative rewards. In turn, this optimal policy empowers the agent to accurately identify and extract the key emotional segments embedded within the EEG signals (Fig. 1).

3.2. Distributed prototype estimation

To effectively capture the emotional distribution embedded in the EEG signals, we introduce a distributed prototype estimation framework. In this framework, each emotion is modeled as a prototype, which facilitates the representation of the emotional landscape and guides the reinforcement learning process through a reward function informed by global emotional characteristics. From a neuroscience perspective, Aydın and Onbaşı [31] leveraged the discrete emotion model hypothesis to successfully differentiate fear and anger through music-evoked brain activity patterns. Building on this foundation, Kılıç and Aydın [32] further demonstrated the link between emotional arousal and brain connectivity, offering a robust framework for EEG-based emotion recognition. In this work, we leverage the efficiency of the K-Means clustering algorithm to derive robust emotion prototypes under the discrete emotion hypothesis.

Specifically, we leverage the heuristic efficiency of the K-Means clustering algorithm to derive robust emotion prototypes. By feeding the differential entropy (DE) features from all subjects' EEG data into K-Means, we obtain a set of cluster centers $\{C^i\}_{i=1}^{n_{clusters}}$, where each C^i corresponds to an emotion category and $n_{clusters}$ is the number of emotion categories in the given database. Each cluster center, or prototype, is computed by averaging the features of all samples belonging to that emotion. Later, these prototypes would be incorporated into the reward function, ensuring that the reinforcement learning process is guided by a comprehensive, globally-informed representation of the emotional distribution captured in the EEG data. Formally, the prototype feature

vector for the C^i emotion category is defined as:

$$\mu_c = \frac{1}{|C^i|} \sum_{x_i \in C^i} f(x_i), \quad (1)$$

where $C^i = (x_i, y_i = c)_{i=1}^{n_{clusters}}$ represents the set of samples labeled with emotion C^i , and $|C^i|$ denotes the number of samples in this category.

The K-Means algorithm iteratively assigns data points to clusters and updates the cluster centers to minimize the sum of squared Euclidean distances between the points and their corresponding centroids:

$$\arg \min_C \sum_{i=1}^{n_{clusters}} \sum_{x \in C^i} \|x - \mu_c\|^2. \quad (2)$$

This objective ensures that the cluster centers are optimally positioned as representative prototypes of the underlying emotional distribution.

In addition to the mean, it is essential to estimate the distribution of the data to fully capture the underlying variability of each emotion category. We quantify the intra-cluster variance of i th cluster, σ_i^2 , which is computed as the mean of the squared distances between each data point and its cluster center as

$$\sigma_i^2 = \frac{1}{|C^i|} \sum_{x_i \in C^i} \|x_i - \mu_c\|^2, \quad (3)$$

This variance metric reveals the degree of dispersion in the EEG features, highlighting the individual variability within each emotion. By examining the distribution rather than relying solely on the mean, we gain a more nuanced understanding of the complexity and heterogeneity inherent in emotional responses. In the implementation, the number of the clusters ($n_{clusters}$) is equal to the number of emotion categories in the given database.

3.3. Distribution-prototype reward

To effectively leverage the rich emotional information embedded in EEG signals, we introduce a novel distribution-prototype reward learning framework. By leveraging global distribution data obtained through distributed prototype estimation, our method infuses the reinforcement learning model with a comprehensive understanding of emotion dynamics. This integration of global emotion distribution into the reward function not only enhances the Actor's ability to discern key segments that closely align with the target emotion but also provides a robust mechanism for handling the inherent variability and non-stationarity of EEG signals.

From a probabilistic perspective, our framework leverages global emotion distribution information derived from emotion prototypes. Specifically, we design two complementary reward functions, namely the center reward and the inter-intra reward, which jointly assess the intensity of the emotional signal and the confidence in a sample's category membership.

Center Reward. The center reward quantifies the intensity of emotional expression in the current EEG sample by measuring its proximity to the corresponding cluster center as

$$Reward_{center} = \frac{1}{1 + dist}, \quad (4)$$

where $dist$ is defined as

$$dist = ||x_{ci} - \mu_{ci}||_2. \quad (5)$$

Here, $dist$ reflects how close a sample is to its category's prototype, thereby indicating the strength of the emotional expression. This measure also helps mitigate the impact of noise and outliers resulting from the non-stationarity of EEG signals.

Inter-Intra Reward. To evaluate the confidence that an EEG sample belongs to a specific emotional category, we introduce the inter-intra reward function from a probabilistic perspective. This function employs inverse variance weighting to compute two components: intra-category distance (intra) and a weighted distance from the centroids of other categories (inter) as

$$Reward_{inter-intra} = \exp\left(-\frac{Intra}{Inter}\right), \quad (6)$$

where

$$Intra = ||x_{ci} - \mu_{ci}||_2, \quad (7)$$

$$Inter = \frac{\sum_{x_i \notin C^i} \frac{||x_i - x_{ci}||_2}{\frac{1}{\sigma_i^2}}}{\sum_{x_i \notin C^i} \frac{1}{\sigma_i^2}}, \quad (8)$$

The inter-intra reward ensures that potential key emotional segments close to the target prototype and far from other category prototypes receive a higher reward, thereby guiding the Actor to maximize long-term expected reward.

Adaptive Fusion via Inverse Variance Weighting. To further enhance the integration of similarity information between the current EEG features and prototypes from other categories, we adopt an inverse variance weighting strategy for adaptive fusion. This approach systematically combines multiple similarity metrics, primarily quantified by Euclidean distances, to yield a more robust estimation of inter-category relationships.

The theoretical foundation for this strategy is provided by the concept of Inverse Variance Weighting [33]. Given a series of independent measurements, y_i with variances σ_i^2 , the inverse variance weighted average is defined as

$$\hat{y} = \frac{\sum_i y_i / \sigma_i^2}{\sum_i 1 / \sigma_i^2}. \quad (9)$$

Among all methods of weighted averaging, the inverse variance weighted average exhibits the least variance. Its variance is expressed as

$$D^2(\hat{y}) = \frac{1}{\sum_i 1 / \sigma_i^2}. \quad (10)$$

When the variances of the measurements are equal, the inverse variance weighted average is equivalent to the simple arithmetic average.

3.4. Optimization process

Our model is trained using Proximal Policy Optimization (PPO) [34], a policy-based Actor-Critic algorithm that efficiently searches for emotionally relevant EEG key segments at the trial level. In this framework,

the model's Actor interacts with the environment to iteratively explore and learn emotionally prototypical policies, while the Critic estimates state values to guide policy improvements. To improve sample efficiency during on-policy training, we introduce importance sampling. This allows us to reuse trajectories by defining the probability ratio between the current and previous policies as follows:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}, \quad (11)$$

where $r_t(\theta)$ quantifies the change in policy.

For more accurate cumulative return estimation, we further incorporate Generalized Advantage Estimation (GAE). First, we compute the temporal-difference error:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad (12)$$

which is derived from the Bellman equation [35] with $V(s)$ representing the state value function learned by the Critic. GAE then recursively aggregates these errors to estimate the advantage:

$$\hat{A}_t^{GAE(\gamma, \lambda)} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^V = \delta_t^V + \gamma \lambda \hat{A}_{t+1}^{GAE(\gamma, \lambda)}, \quad (13)$$

where γ is the discount factor and λ is a hyperparameter that balances bias and variance.

To further stabilize training and avoid excessive policy updates, we apply the PPO-Clip mechanism. By constraining the ratio between the new and old policies, PPO-Clip reduces oscillations and accelerates convergence. The clipped objective function is expressed as:

$$\mathcal{L}_{policy}(\theta) = \hat{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (14)$$

where ϵ sets the boundaries for acceptable policy changes.

The Critic network, responsible for estimating $V(S_t)$, predicts the true discounted cumulative reward. Its performance is optimized by minimizing the mean squared error (MSE) between its predicted value and the actual reward:

$$L(\theta) = \mathbb{E} \left[(V(s_t) - R_t)^2 \right], \quad (15)$$

where R_t represents the discounted cumulative reward at time t . In summary, the training process integrates two synergistic components. The Actor continually interacts with the emotional space to refine its decision-making strategy based on environmental rewards, while the Critic provides essential feedback by estimating state values, collectively steering the model toward optimal performance.

3.5. Key segment extraction via regularization and sequential inference

Our approach to key segment extraction integrates both controlled selection and sequential inference into a unified framework. The primary objective is to extract key emotional segments while ensuring that the number of selected key segments remains within a desired range.

Regularization for Controlled Selection. To prevent the Actor from selecting an excessive number of key segments in an episode, we introduce a regularization term that constrains the average action probability of the learned policy function. The regularization loss is defined as:

$$L_{prob} = ||\frac{1}{T} \sum_{t=1}^T p_t - \alpha||^2, \quad (16)$$

where α represents the desired proportion of key emotional segments. The parameters θ of the policy function are then updated using the Adam optimizer according to:

$$\theta = \theta - \varphi \nabla_\theta (-\mathcal{L}_{policy} + \beta \mathcal{L}_{prob}), \quad (17)$$

with φ denoting the learning rate and β as the regularization coefficient.

Table 1

Detailed description of the SEED, SEED-IV and DEAP datasets.

Datasets	Subject	Sessions	Trials	Channels	Sampling rate (Hz)	Classes
SEED	15	3	15	62	1000	3 (Negative, Neutral, Positive)
SEED-IV	15	3	24	62	1000	4 (Happy, Neutral, Sad, Fear)
DEAP	32	1	40	32	512	2 Valence (Negative, Positive) 2 Arousal (Calm, Active)

Sequential Inference for Key Segment Identification. Algorithm 1 trains the model on the training set to obtain the parameters θ . More detailed descriptions of Dynamic Emotion Modeling are presented in the Supplementary Materials Appendix D. During inference on the test dataset, the RL-trained model predicts sample-level probabilities for the sequence $s_{t=1}^T$, effectively serving as importance scores for each frame. The reward function, which quantifies potential emotional intensity via a distance metric analysis, guides this probabilistic weighting process. Frames are then ranked in descending order of probability, and experimental observations indicate that the most salient frames tend to maintain temporal continuity, suggesting an intrinsic link between emotional intensity progression and the chronological structure of the sequence.

After ranking, the top 40 % of segments are selected as key moments. These key segments are subsequently analyzed using both traditional machine learning methods and deep learning techniques, with a multi-layer perceptron (MLP) serving as a representative example.

Algorithm 1 The training process of Emotion Agent.

Input: DE feature sequence $\{s_t\}_{t=1}^T$ from training set

Output: Parameters θ of the Emotion Agent.

- 1: Input All Subjects DE feature data into K-Mean, calculate μ_i according to Eq. (2), calculate σ^2 according to Eq. (4);
 - 2: Initial actor parameters θ_0 , initial critic parameters ϕ_0 ;
 - 3: **for** $i \rightarrow 1, 2, \dots, \xi$ **do**
 - 4: Input a DE feature sequence $\{s_t\}_{t=1}^T$ from training set
 - 5: Collect set of trajectories $D_k = \{\tau_i\}$ by policy π_k of actor;
 - 6: Compute generalized advantage estimates $\hat{A}_t$, based on the current value function V_{ϕ_k} ;
 - 7: Update the policy of actor by maximizing the PPO-Clip objective function Eq. (11);
 - 8: Update the value function of critic by regression on Mean-Squared Error;
 - 9: **end for**
 - 10: Save the parameters θ of the Emotion Agent.
-

4. Experiments

4.1. Datasets

We conduct experiments on three publicly available EEG datasets: SEED [36], SEED-IV [37], and DEAP [38]. These datasets are widely used as benchmarks for evaluating EEG-based emotion recognition algorithms. The dataset statistics are summarized in Table 1.

SEED EEG signals were recorded from 15 subjects (7 males and 8 females) using a 62-channel ESI NeuroScan system following the international 10–20 standard. The raw EEG data were sampled at 1000 Hz. Each participant completed three experimental sessions. In every session, they viewed 15 film clips designed to elicit emotional responses. These clips were categorized into three emotional states: positive, neutral, and negative, with five clips per category.

SEED-IV EEG signals were recorded from 15 subjects (7 males and 8 females) using a 62-channel ESI NeuroScan system along with an SMI eye tracker. Each subject participated in three experimental sessions. In each session, they watched 24 film clips designed to elicit four emotional

states: happy, sad, fearful, and neutral. Each emotional state contained six film clips.

DEAP EEG signals were recorded from 32 subjects (16 males and 16 females) using a 32-channel Biosemi Active Two system at a sampling rate of 512 Hz. Each participant viewed 40 one-minute-long music videos. After each video, participants rated their emotional states on four scales: Valence, Arousal, Dominance, and Liking. These ratings were binarized using a predefined threshold to obtain discrete emotion labels.

In line with previous research [6,39], Differential Entropy (DE) features are extracted from each 1-second EEG segment and used as input to the model.

4.2. Implementation details

During the reinforcement learning phase, we operate on trial-level EEG data. Each training sample consists of the DE features corresponding to one trial from a given subject. This process is modeled as a sequential decision-making task, where the agent performs a series of binary actions to identify emotion-relevant EEG segments. The discrete action space is defined as $A = \{0, 1\}$, indicating whether to select a given EEG segment. The actor network consists of one LSTM layer with 128 hidden units, followed by two fully connected (FC) layers with dimensions $256 \rightarrow 128$ and $128 \rightarrow 2$. The critic network has a similar structure, comprising one LSTM layer with 128 units and two FC layers with dimensions $256 \rightarrow 128$ and $128 \rightarrow 1$. Both networks are trained using the Adam optimizer. The learning rate for the actor is set to $3e-3$, while the critic is optimized with a learning rate of $1e-2$.

We employ the Proximal Policy Optimization (PPO) algorithm for policy learning. The hyperparameters are configured as follows: the discount factor γ is set to 0.98, the Generalized Advantage Estimation (GAE) parameter λ is set to 0.95, and the clipping threshold ϵ is set to 0.2 (refer to Eqs. 11, 13, and 14). All experiments are implemented in PyTorch 1.13.1 and conducted on a machine equipped with an NVIDIA GeForce RTX 3090 GPU.

To evaluate the performance of the proposed method, we adopt two experimental protocols. The first is the Cross-Subject setting, where we use subject-independent, subject-level leave-one-out cross-validation (LOOCV) to assess the generalization ability of the model across unseen individuals. The second is the Within-Subject setting, which employs video-level LOOCV to measure performance consistency within individual subjects. We report two key evaluation metrics: Accuracy and F1-Scores, both of which reflect the quality and effectiveness of the extracted emotional segments in improving recognition performance.

4.3. Experimental results

To evaluate the effectiveness of our proposed Emotion Agent, we conduct leave-one-out cross-validation (LOOCV) on the SEED dataset under consistent experimental protocols. As shown in Table 2, the results demonstrate that our method achieves an accuracy of 61.93 %, representing a 3.71 % improvement over the state-of-the-art unsupervised method TAS-Net. Unlike supervised approaches that benefit from label information but often suffer from overfitting and sensitivity to label noise, our Emotion Agent adopts a fully unsupervised reinforcement learning framework. By integrating a prototype-guided reward design and temporal decision-making strategy, it selectively extracts emotionally relevant EEG segments without relying on ground truth annotations.

Table 2

Model performance (%) for cross-subject emotion recognition on the SEED dataset.

Methods	P_{acc}	P_f	Methods	P_{acc}	P_f
<i>Supervised</i>			<i>Within-subject cross-validation</i>		
DGCNN [40]	90.40	89.27	BiDANN [41]	92.38	91.32
R2G-STNN [42]	93.38	90.25	RGNN [43]	94.24	92.37
<i>Supervised</i>			<i>Subject-level LOOCV</i>		
DANN [44]*	80.65	80.29	DDC [45]*	81.18	80.84
BIDANN [41]	83.28	–	RGNN [43]	85.30	–
DResNet [46]	85.30	–	JDA [47]	88.28	–
<i>Unsupervised</i>			<i>Subject-level LOOCV</i>		
Liang [48] et al. *	42.32	50.36	DSN [12]*	53.22	51.93
EEGFuseNet [49]	40.09	42.33	TAS-Net [11]	58.22	53.14
Emotion agent (Ours)	61.93	60.01			

Table 3

Model performance (%) for cross-subject emotion recognition on the SEED-IV dataset.

Methods	P_{acc}	P_f	Methods	P_{acc}	P_f
<i>Supervised</i>			<i>Traditional machine learning methods</i>		
KNN	42.74	32.13	KPCA	29.25	28.67
SVM	58.23	46.21	SA	34.74	31.52
TCA	44.11	45.79	CORAL	48.14	51.23
GFK	43.10	37.56	RF	52.67	42.24
<i>Supervised</i>			<i>Deep learning methods</i>		
DAN [50]*	54.87	46.07	DANN [47]*	57.63	45.29
DCORAL [51]*	62.77	61.25	DDC [40]*	63.69	60.91
MS-MDA [45]	57.36	–	MMD [52]	59.34	–
EEGNet [53]*	53.96	40.73	Emotion agent (Ours)	64.66	56.71
<i>Unsupervised</i>			<i>Deep learning methods</i>		
Liang et al. [48] *	33.27	28.13	DSN [12]*	30.09	27.68
TAS-Net [11]*	39.72	32.24	Emotion agent (Ours)	43.41	29.75

We further evaluate the generalizability of the Emotion Agent on the SEED-IV dataset, which includes four distinct emotional categories. As shown in Table 3, our method first extracts emotionally informative EEG segments, which are then classified using a multilayer perceptron (MLP) composed solely of fully connected layers. This approach achieves an accuracy of 64.66 %, outperforming both traditional machine learning algorithms and most deep learning models. In addition to supervised evaluation, we also assess the method's unsupervised performance. By applying K-Means to the extracted keyframes under a cross-subject LOOCV setting, our method yields a 3.69 % improvement over the current unsupervised state-of-the-art, TAS-Net. Unlike conventional supervised models that indiscriminately process all EEG segments, risking the inclusion of noisy or irrelevant data, our Emotion Agent selectively identifies emotionally salient information, leading to more consistent and robust recognition across emotional categories.

We also verify the Emotion Agent on the DEAP dataset, which involves both Arousal and Valence prediction tasks. As reported in Table 4, our method continues to exhibit strong performance under subject-independent LOOCV settings. For Arousal classification, it outperforms the state-of-the-art ATDD-LSTM model by 11.33 %. For binary Valence classification, it surpasses the DGCNN baseline by 5.17 %, demonstrating the Emotion Agent's effectiveness in capturing subtle emotional variations in EEG signals. We further examine its performance in the unsupervised setting by applying K-Means clustering to the extracted keyframes. Compared to TAS-Net, our method achieves a 7.19 % improvement in Arousal classification and a 1.42 % gain in Valence classification. These results confirm that the Emotion Agent can

Table 4

Model performance (%) for cross-subject emotion recognition on the DEAP dataset.

Methods	Arousal		Valence	
	P_{acc}	P_f	P_{acc}	P_f
<i>Supervised</i>		<i>Subject-dependent</i>	<i>Video-level LOOCV</i>	
DT-CWPT [54]	64.30	–	66.20	–
EMD [55]	71.99	–	69.10	–
<i>Supervised</i>		<i>Subject-independent</i>	<i>Subject-level LOOCV</i>	
DGCNN [40]	61.10	–	59.29	–
ATDD-LSTM [48]	72.97	–	69.06	–
DANN*	70.14	52.99	64.01	40.96
DDC*	55.35	37.68	61.95	39.39
DCORAL*	59.75	51.36	62.50	41.28
Emotion agent (Ours)	84.30	53.28	64.46	41.56
<i>Unsupervised</i>		<i>Subject-dependent</i>	<i>Video-level LOOCV</i>	
Liang et al. * [49]	62.34	60.44	56.25	61.75
<i>Unsupervised</i>		<i>Subject-independent</i>	<i>Subject-level LOOCV</i>	
DSN* [49]	58.55	42.28	56.44	43.29
TAS-Net* [11]	69.53	45.64	56.84	43.43
Emotion-agent (Ours)	76.72	50.19	58.26	46.54

reliably extract emotionally relevant EEG segments and enhance emotion recognition performance across both supervised and unsupervised paradigms.

Overall, experiments on the SEED, SEED-IV, and DEAP datasets demonstrate that the Emotion Agent effectively isolates the key EEG segments closely related to emotional states. By mitigating issues such as label noise and redundant data with its designed reward function, our method significantly enhances emotion recognition performance for both discrete and dimensional emotional assessments.

5. Discussion

5.1. Emotion agent comparative evaluation

To evaluate the performance of our proposed Emotion Agent in enhancing the accuracy and reliability of emotion recognition, we conduct comparative experiments on three widely used datasets: SEED, SEED-IV, and DEAP. In these experiments, we compare two scenarios. In the first, the Emotion Agent extracts key emotion-related EEG segments prior to classification; in the second, raw EEG signals are directly fed into the classifiers. For classification, we utilize nine methods (1) SVM, (2) KNN, (3) K-Means, (4) MLP, (5) EEGNet, (6) DAN, (7) DANN, (8) DDC, and (9) DCORAL.

Table 5 presents the three-class classification results on the SEED dataset using a leave-one-subject-out cross-validation approach. For example, when the SVM classifier is applied directly to the raw EEG data, we obtain an accuracy of 64.47 %, a precision of 69.54 %, a recall of 64.33 %, and an F1-score of 62.39 %. In contrast, after applying the Emotion Agent to extract emotion-related EEG segments, the performance improves to 74.47 % accuracy (an increase of 9.65 %), 76.54 % precision (an increase of 7.37 %), 73.93 % recall (an increase of 9.60 %), and a 10.8 % boost in F1-score. Overall, the accuracy on the SEED dataset increases by an average of 11.07 %.

For the SEED-IV dataset with four classes, based on leave-one-subject-out cross-validation, the comparative evaluation of using and not using the proposed Emotion Agent is reported in Table 6. The results show that the Emotion Agent approach yields an average accuracy improvement of 9.88 %, with statistical significance confirmed by the paired *t*-test (Fig. 2(b), $p < 0.05$).

In addition to demonstrating benefits in discrete emotion recognition, we evaluate the performance of the Emotion Agent on dimensional

Table 5

Emotion recognition performance (%) with the traditional classification methods using subject-independent LOOCV strategy on SEED dataset, under the conditions without (w/o) and with (w) the proposed Emotion Agent method.

Methods	Sampling	SEED three-class			
		Accuracy	Precision	Recall	F1-Scores
SVM	w/o	64.47 ± 11.16	69.54 ± 9.53	64.33 ± 11.30	62.39 ± 12.15
	w	74.12 ± 09.34	76.91 ± 08.88	73.93 ± 09.42	73.19 ± 09.98
KNN	w/o	55.44 ± 08.67	57.17 ± 07.40	55.16 ± 08.57	54.11 ± 08.61
	w	64.37 ± 09.79	65.43 ± 10.17	64.17 ± 09.68	63.35 ± 10.80
K-Means	w/o	42.07 ± 10.59	25.61 ± 10.60	41.44 ± 10.71	30.55 ± 11.15
	w	61.93 ± 14.13	61.79 ± 15.03	61.59 ± 14.17	60.01 ± 14.28
MLP	w/o	68.31 ± 11.01	72.05 ± 10.16	68.20 ± 10.92	67.18 ± 11.42
	w	78.14 ± 06.15	80.07 ± 06.07	77.91 ± 06.28	77.19 ± 06.62
EEGNet	w/o	69.01 ± 07.38	72.98 ± 09.23	68.94 ± 07.42	67.88 ± 08.86
	w	80.24 ± 06.48	81.58 ± 05.63	79.04 ± 06.61	78.68 ± 06.67
DAN	w/o	83.81 ± 08.56	84.26 ± 06.72	82.10 ± 07.39	81.72 ± 07.73
	w	90.33 ± 07.12	89.26 ± 05.39	90.10 ± 06.39	90.72 ± 07.73
DANN	w/o	80.65 ± 11.01	85.09 ± 09.92	85.65 ± 11.23	80.98 ± 10.19
	w	91.22 ± 07.34	92.79 ± 04.07	91.14 ± 07.75	91.14 ± 07.71
DDC	w/o	81.18 ± 05.43	82.26 ± 05.60	81.10 ± 05.39	80.84 ± 05.59
	w	92.04 ± 05.37	92.89 ± 04.93	91.96 ± 04.93	91.96 ± 05.22
DCORAL	w/o	78.35 ± 06.09	81.89 ± 04.24	78.26 ± 06.19	78.05 ± 06.33
	w	90.55 ± 06.28	91.99 ± 05.36	90.43 ± 06.36	90.27 ± 06.55
Average improvement		↑ 11.07	↑ 11.26	↑ 11.97	↑ 12.60

Table 6

Emotion recognition performance (%) with the traditional classification methods using subject-independent LOOCV strategy on SEED-IV dataset, under the conditions without (w/o) and with (w) the proposed Emotion Agent method.

Methods	Sampling	SEED-IV four-class			
		Accuracy	Precision	Recall	F1-scores
SVM	w/o	49.50 ± 11.96	51.28 ± 13.77	50.05 ± 12.11	46.21 ± 13.01
	w	58.23 ± 09.72	59.90 ± 13.04	59.25 ± 09.11	53.33 ± 10.88
KNN	w/o	34.24 ± 08.52	34.25 ± 07.12	34.44 ± 08.69	32.13 ± 07.95
	w	42.74 ± 10.48	36.24 ± 12.99	36.64 ± 11.96	34.52 ± 12.14
K-Means	w/o	33.76 ± 05.89	16.76 ± 06.82	31.00 ± 05.31	20.30 ± 06.17
	w	43.41 ± 08.10	15.84 ± 09.18	29.30 ± 06.77	29.75 ± 08.67
MLP	w/o	52.04 ± 11.32	54.75 ± 13.53	53.72 ± 10.89	49.05 ± 11.43
	w	64.66 ± 11.77	58.47 ± 15.19	61.46 ± 13.10	56.71 ± 13.86
EEGNet	w/o	53.96 ± 08.72	48.83 ± 08.96	47.12 ± 08.96	40.73 ± 11.23
	w	65.95 ± 06.95	51.65 ± 12.52	56.33 ± 05.71	53.05 ± 06.93
DAN	w/o	54.87 ± 08.31	46.21 ± 08.67	53.29 ± 05.35	46.07 ± 07.73
	w	63.75 ± 10.47	59.18 ± 12.48	57.14 ± 08.83	51.60 ± 12.38
DANN	w/o	57.63 ± 07.92	58.42 ± 10.56	52.93 ± 07.19	45.29 ± 09.49
	w	66.97 ± 07.34	64.74 ± 12.07	64.66 ± 07.75	62.39 ± 08.71
DDC	w/o	63.69 ± 06.36	78.98 ± 06.64	62.02 ± 06.86	60.91 ± 07.04
	w	71.80 ± 08.39	72.05 ± 09.38	64.68 ± 09.55	64.22 ± 08.83
DCORAL	w/o	62.77 ± 08.64	71.17 ± 06.64	63.89 ± 08.35	61.25 ± 09.73
	w	73.27 ± 08.43	77.30 ± 07.55	75.77 ± 08.12	73.03 ± 09.95
Average improvement		↑ 09.81	↑ 08.84	↑ 06.31	↑ 07.41

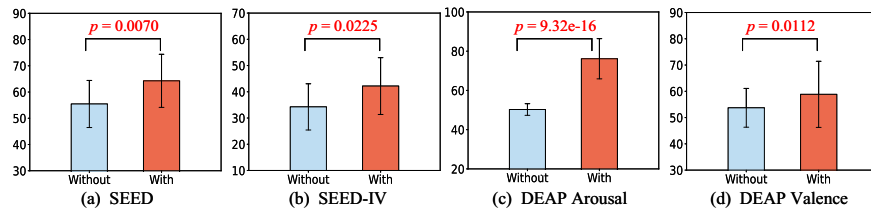


Fig. 2. The obtained p-values of the paired t -test between the proposed emotion agent and without in terms of P_{acc} . The error bars represent the standard deviation of the obtained experimental results.

emotion recognition. On the DEAP dataset, we perform two-class classifications for both arousal and valence dimensions (Tables 7 and 8). The application of the Emotion Agent yields an average accuracy improvement of 27.54 % for arousal classification and 6.29 % for valence classification.

To validate the statistical significance of these improvements, we performed paired t -tests, with results presented in Fig. 2. The performance

gains achieved by Emotion Agent are statistically significant ($p < 0.05$) across datasets and classifier configurations, reinforcing the reliability of our method. For the SEED dataset, the paired t -test results (Fig. 2(a)) confirm that these improvements are statistically significant ($p < 0.05$). For the SEED-IV dataset, the paired t -test results (Fig. 2(b)) confirm that these improvements are statistically significant ($p < 0.05$). For the DEAP dataset (Arousal and Valence), Fig. 2(c) and

Table 7

Emotion recognition performance (%) with the traditional classification methods using subject-independent LOOCV strategy on DEAP Arousal dataset, under the conditions without (w/o) and with (w) the proposed Emotion Agent method.

Methods	Sampling	DEAP arousal two-class			
		Accuracy	Precision	Recall	F1-scores
SVM	w/o	49.12 ± 07.96	51.10 ± 06.46	50.72 ± 05.46	45.83 ± 08.26
	w	84.30 ± 12.53	49.96 ± 22.17	57.81 ± 18.15	53.28 ± 20.40
KNN	w/o	50.27 ± 02.92	49.70 ± 02.11	49.72 ± 02.09	48.27 ± 02.67
	w	76.72 ± 08.46	50.19 ± 02.87	49.51 ± 03.18	48.28 ± 02.82
K-Means	w/o	55.24 ± 09.28	27.62 ± 04.64	50.00 ± 00.00	35.36 ± 03.82
	w	84.30 ± 12.53	49.96 ± 22.17	57.81 ± 18.15	53.28 ± 20.40
MLP	w/o	64.84 ± 00.92	64.84 ± 00.92	64.84 ± 00.92	64.83 ± 00.92
	w	84.30 ± 12.53	50.61 ± 22.17	57.81 ± 18.15	53.28 ± 20.40
EEGNet	w/o	55.24 ± 09.06	45.23 ± 08.83	50.01 ± 00.24	45.67 ± 04.04
	w	83.29 ± 09.35	55.40 ± 05.18	50.00 ± 00.00	48.23 ± 03.60
DAN	w/o	57.63 ± 07.92	58.42 ± 10.56	52.93 ± 07.19	45.29 ± 09.49
	w	67.75 ± 07.34	59.18 ± 12.07	57.14 ± 07.75	51.60 ± 08.71
DANN	w/o	70.14 ± 05.34	71.79 ± 05.99	60.91 ± 03.51	52.99 ± 04.14
	w	87.56 ± 09.34	68.30 ± 10.07	58.14 ± 04.57	57.03 ± 06.71
DDC	w/o	55.35 ± 09.12	45.87 ± 14.05	54.03 ± 00.17	37.68 ± 03.92
	w	79.80 ± 06.35	58.40 ± 05.18	60.00 ± 0.00	51.23 ± 03.60
DCORAL	w/o	59.75 ± 07.71	64.82 ± 10.97	55.11 ± 05.18	53.36 ± 07.50
	w	86.44 ± 08.03	73.91 ± 10.28	54.03 ± 03.99	51.37 ± 06.39
Average improvement		↑ 26.20	↑ 04.46	↑ 02.83	↑ 04.83

Table 8

Emotion recognition performance (%) with the traditional classification methods using subject-independent LOOCV strategy on DEAP Valence dataset, under the conditions without (w/o) and with (w) the proposed Emotion Agent method.

Methods	Sampling	DEAP valence two-class			
		Accuracy	Precision	Recall	F1-scores
SVM	w/o	51.04 ± 06.75	50.86 ± 05.77	51.12 ± 03.92	46.85 ± 06.23
	w	58.27 ± 13.35	51.95 ± 06.92	51.86 ± 04.75	46.72 ± 08.06
KNN	w/o	53.02 ± 06.30	50.20 ± 02.80	50.41 ± 02.10	46.86 ± 04.10
	w	58.26 ± 11.62	49.91 ± 02.90	49.40 ± 03.89	46.54 ± 05.49
K-Means	w/o	56.94 ± 15.52	28.47 ± 07.76	50.00 ± 00.00	35.62 ± 06.73
	w	65.39 ± 19.80	34.26 ± 15.09	51.56 ± 08.70	40.12 ± 13.46
MLP	w/o	60.23 ± 00.15	33.75 ± 00.83	50.00 ± 00.12	40.30 ± 07.11
	w	64.46 ± 19.29	42.18 ± 19.00	51.80 ± 08.75	41.56 ± 14.00
EEGNet	w/o	56.55 ± 08.50	45.93 ± 10.43	50.21 ± 01.03	43.12 ± 07.92
	w	64.77 ± 10.99	52.94 ± 09.09	51.37 ± 02.87	42.85 ± 09.09
DAN	w/o	57.63 ± 07.92	58.42 ± 10.56	56.41 ± 01.29	39.72 ± 07.65
	w	67.75 ± 07.34	59.18 ± 12.07	57.14 ± 07.75	51.60 ± 08.71
DANN	w/o	64.01 ± 09.92	65.52 ± 09.13	54.56 ± 03.73	40.96 ± 03.65
	w	69.23 ± 09.15	65.24 ± 09.13	60.11 ± 05.04	60.36 ± 05.41
DDC	w/o	61.95 ± 11.23	71.17 ± 10.40	58.26 ± 02.50	39.39 ± 04.46
	w	67.75 ± 07.34	59.18 ± 12.07	57.14 ± 07.75	51.60 ± 08.71
DCORAL	w/o	62.50 ± 08.77	63.27 ± 08.58	55.05 ± 03.25	41.28 ± 03.34
	w	71.01 ± 08.97	72.22 ± 08.57	61.27 ± 05.56	58.93 ± 06.18
Average improvement		↑ 6.73	↑ 03.54	↑ 02.13	↑ 07.50

(d) further confirm that these improvements are statistically significant ($p < 0.05$).

In summary, our evaluation shows that the Emotion Agent improves emotion recognition by extracting EEG segments most relevant to emotional states. It consistently boosts performance for both traditional and deep learning classifiers, demonstrating its ability to capture rich emotional features. Moreover, its reward function reflects emotional intensity and enables autonomous selection of key segments, thereby enhancing downstream task accuracy and reliability.

5.2. Optimization process evaluation

To further assess the model optimization process, two experiments were conducted: one to validate the designed reward function, and the other to explore different hyperparameter configurations of Proximal Policy Optimization (PPO).

First, we evaluated the functionality of the proposed reward function by comparing the Distribution-Prototype Reward introduced in this paper with a variant combining Center Reward and only Inter Reward.

Through cross-subject experiments, we analyzed the trend of the reward curve during model optimization under different reward mechanisms. As illustrated in the left panel of Fig. 5, the red curve represents the

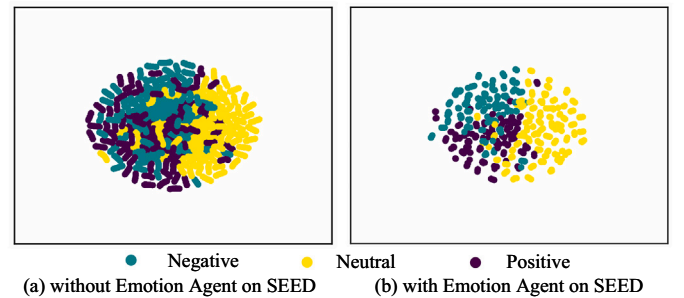


Fig. 3. Comparison of data distribution using t-SNE with and without emotion agent on the SEED dataset.

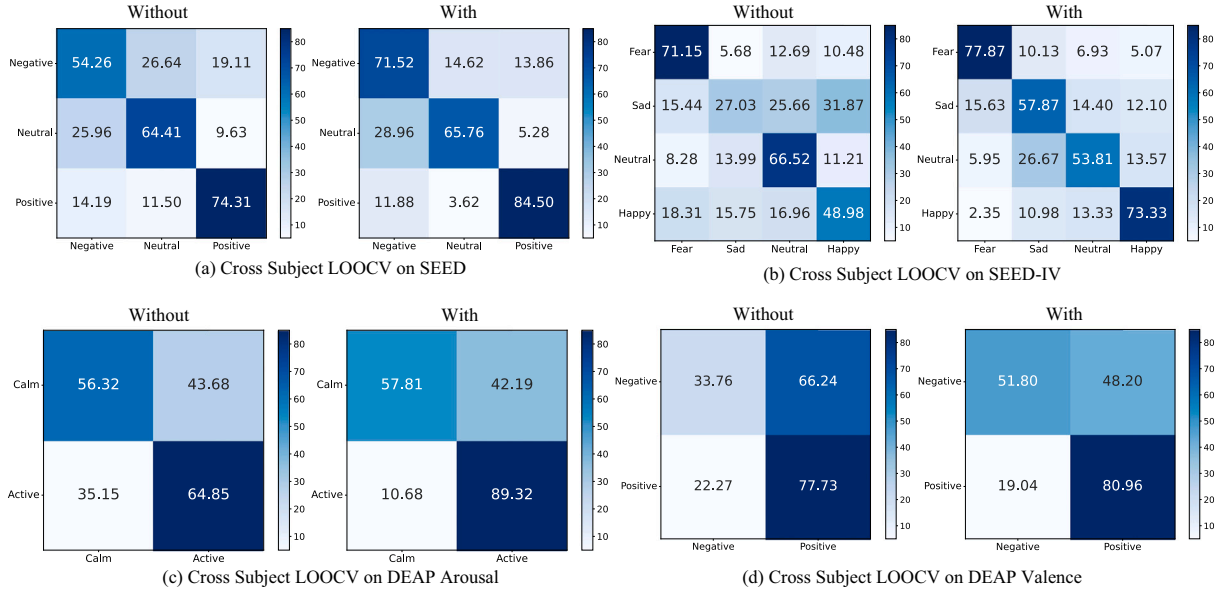


Fig. 4. Comparison of confusion matrices for cross-subject LOOCV emotion recognition with and without emotion agent on the SEED, SEED-IV and DEAP dataset.

results using our complete reward function, while the blue curve corresponds to the combination of Center Reward and Inter Reward alone. Here, Episode denotes the number of times the agent interacts with the environment during training. In each interaction, the agent completes the task of selecting key frames for trial-level EEG signals. Return refers to the cumulative reward obtained in each interaction. The agent acquires optimal strategies through trial-and-error during exploration and exploitation. Notably, the results show that the reward curve using our proposed function converges rapidly, demonstrating that the agent efficiently learns optimal strategies under its guidance, with the curve maintaining an overall upward trend throughout the learning process. In contrast, the blue curve fails to effectively direct the agent's learning: the agent is unable to optimize its strategy via reward feedback, resulting in suboptimal maximization of cumulative expected returns.

Additionally, to explore the EEG emotional space more effectively with our well-designed reward function, we employ PPO to optimize the entire training process. The optimization process curves under different hyperparameters of the PPO algorithm are presented in the right panel of Fig. 5. We perform additional cross-subject experiments on the SEED dataset to analyze the training dynamics and assess the impact of the clipping parameter, ϵ , on model performance. With a carefully designed reward function, the cumulative benefits from extracting key EEG segments steadily increase, and the agent gradually adopts an optimal action strategy. The right panel shows that a lower clipping threshold stabilizes the training process, allowing the agent to learn superior policy. By constraining the ratio between new and old policies, the model converges more steadily and efficiently.

5.3. t-SNE visualization and confusion matrix

We further explore the EEG key segments extracted by our proposed Emotion Agent method. We conduct additional experiments on the SEED dataset and visualize the extracted EEG key fragments with t-SNE, comparing them to the original features. As shown in Fig. 3, it presents a direct comparison between the original DE features and the features obtained after applying the Emotion Agent model for key segment extraction, with t-SNE used for visualization. This comparison clearly demonstrates that the extracted key segments yield a more distinctly separated data distribution, which directly supports the observed improvement in classification accuracy.

In addition, we analyze the confusion matrices for classifications performed with and without the proposed Emotion Agent on the SEED,

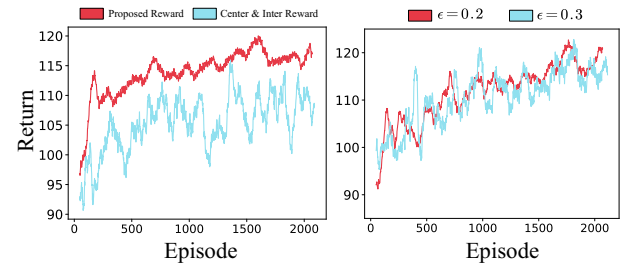


Fig. 5. Evaluation on optimization process.

SEED-IV and DEAP datasets, as shown in Fig. 4. For the SEED dataset, Without the Emotion Agent, the diagonal values, which indicate the correct predictions for the negative, neutral, and positive classes, are 54.26 %, 64.41 %, and 74.31 %, respectively. When the Emotion Agent is incorporated, these values increase significantly to 71.52 %, 65.76 %, and 84.50 %, respectively, and misclassifications are markedly reduced. For the SEED-IV dataset, Without the Emotion Agent, the diagonal values, which indicate the correct predictions for the fear, sad, neutral and happy classes, are 71.15 %, 27.03 %, 66.52 % and 48.98 % respectively. When the Emotion Agent is incorporated, these values increase significantly to 77.87 %, 57.87 %, 53.81 % and 73.33 %. For the DEAP dataset Arousal, Without the Emotion Agent, the diagonal values, which indicate the correct predictions for the calm and positive classes, are 56.32 % and 64.85 %, respectively. When the Emotion Agent is incorporated, these values increase significantly to 57.81 % and 89.32 %, respectively, and misclassifications are markedly reduced. For the DEAP dataset Valence, Without the Emotion Agent, the diagonal values, which indicate the correct predictions for the negative, neutral, and positive classes, are 33.76 % and 77.73 %. When the Emotion Agent is incorporated, these values increase significantly to 51.80 %, and 80.96 %. These results clearly demonstrate that integrating the Emotion Agent results in consistently higher classification accuracy across all the emotion categories.

5.4. Visualization of EEG keyframe topology

We further investigate the EEG key segments extracted by the proposed Emotion Agent. The corresponding keyframe video images and

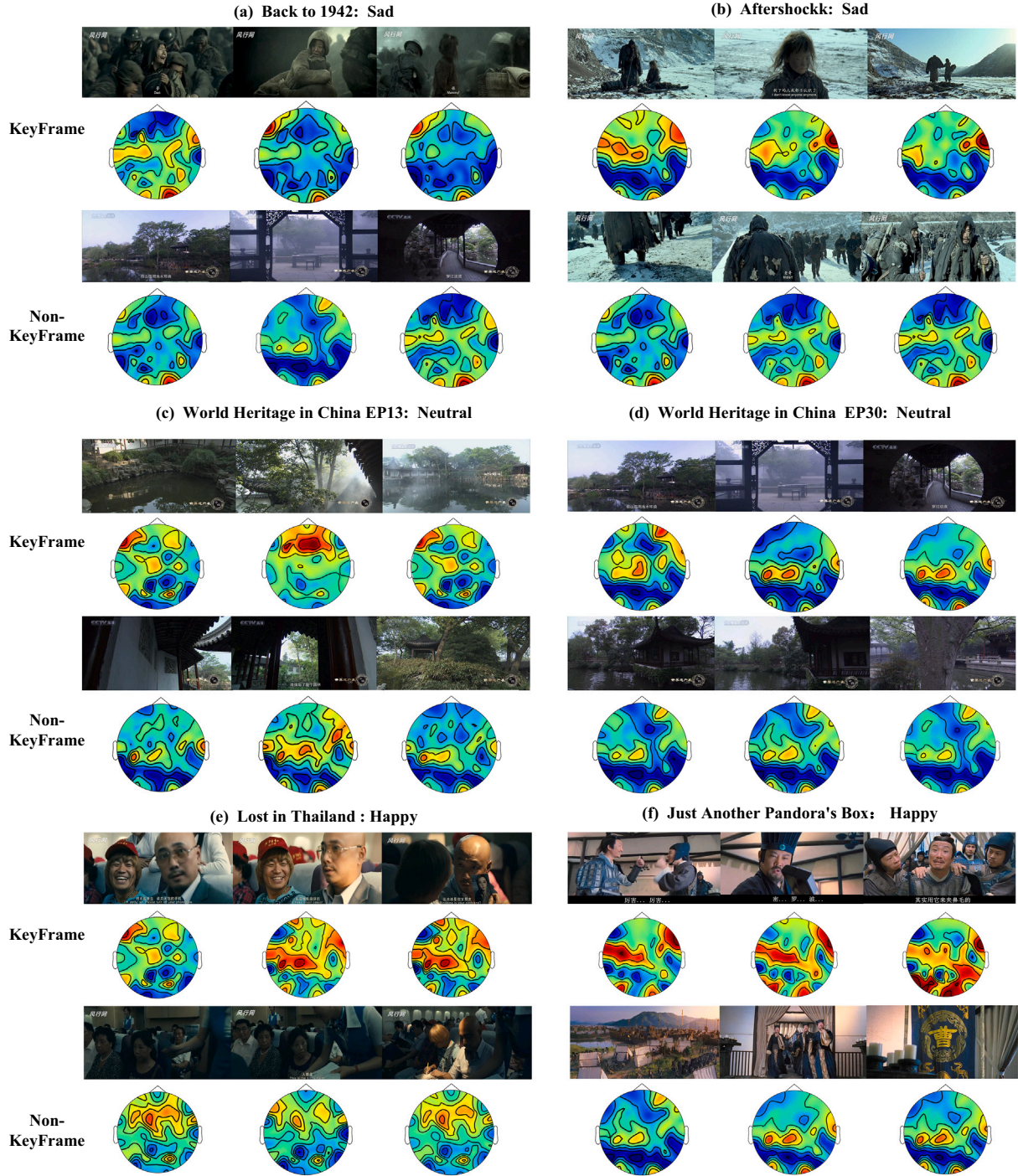


Fig. 6. Comparative visualization of keyframe and non-keyframe brain maps extracted by the proposed model emotion agent under different emotional stimuli. (a) and (b) visualization of keyframe and non-keyframe brain maps under negative emotional stimulation; (c) and (d) visualization of keyframe and non-keyframe brain maps under neutral emotional stimulation; (e) and (f) visualization of keyframe and non-keyframe brain maps of brain region activation under positive emotional stimulation.

brain topographies are visualized. As shown in Fig. 6, comparative analysis of brain activation patterns between keyframes and non-keyframes reveals enhanced activation in regions linked to cognitive-emotional processing. This provides neurophysiological evidence that subjects are likely experiencing intense emotional states during the extracted segments.

6. Conclusion

In this paper, we proposed a novel unsupervised deep reinforcement learning framework, Emotion Agent, to automatically extract key emotional segments from continuous EEG signals. By combining heuristic search with a distribution-prototype-based reward function, our method

effectively filters out irrelevant data and focuses on segments that accurately reflect emotional states. Extensive experiments on multiple benchmark datasets demonstrate that Emotion Agent significantly enhances the accuracy and robustness of emotion recognition, highlighting its potential for real-world aBCI applications.

CRedit authorship contribution statement

Zhihao Zhou: Writing – original draft, Visualization, Validation, Methodology. **Li Zhang:** Investigation, Funding acquisition, Conceptualization. **Qile Liu:** Methodology, Investigation. **Gan Huang:** Visualization, Software, Resources. **Zhuliang Yu:** Supervision, Project administration. **Zhen Liang:** Writing – review & editing, Writing – original draft, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.neucom.2025.130951](https://doi.org/10.1016/j.neucom.2025.130951).

Data availability

All the data have been shared online.

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